AI-1 Spring 26 Project Proposal

Team Members

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Team ID (3)

Idea Short Name (Car pathfinding)

Idea Description

This project implements an AI-controlled car that navigates from a starting position to a target goal using the A* pathfinding algorithm. The environment is represented as a grid where obstacles are automatically detected. The system computes the optimal path and guides the car to avoid obstacles and reach the destination efficiently.

Agent Architecture

☐ Simple Reflex Agent

☐ Model-Based Reflex Agent

√ Goal-Based Agent

☐ Utility-Based Agent

Problem-Solving Framework

√ Classical Search

☐ Local Search (Optimization)

☐ Adversarial Search (Two-Player Games)

☐ Constraint Satisfaction Problem

Proposed Algorithm(s)

A* (A-Star) Pathfinding Algorithm

PEAS

Performance Measures	Shortest path length Avoiding obstacles
Environment	Static obstacles (boxes, barriers)
Actuators	Car movement
Sensors	Grid system

Environment Properties

√ Fully Observable ☐ Partially Observable	√ Deterministic ☐ Stochastic ☐ Strategic	√ Discrete ☐ Continuous
√ Single-Agent ☐ Multi-Agent	√ Sequential ☐ Episodic	☐ Static √ Dynamic